DEVIXO SOLUTIONS

BUSINESS INTELLIGENCE & DASHBOARD DEVELOPMENT

Data Science Internship | Week 3 | Task 03

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Internship Domain	Data Science
Task	03 – Business Intelligence & Dashboard Development
BI Tool	Microsoft Power BI Desktop
Dataset	Simulated Retail Sales Dataset
Submission	November 2026

PROJECT OVERVIEW

This report documents the preparation, design, and interpretation of an interactive retail sales dashboard created in Microsoft Power BI Desktop. The dashboard consolidates key sales and profitability indicators and presents performance across years, product categories, regions, shipping modes, and sub-categories.

Dataset note: The earlier project brief identifies the retail records as simulated data prepared for educational analysis. The results in this report describe that dataset and must not be presented as the actual performance of a real company.

EXECUTIVE SUMMARY

The project turns transaction-level retail data into a visual business intelligence report. The dataset was loaded into Power BI, reviewed and prepared in Power Query, and used to build measures and report visuals. The live dashboard screenshot supplied for this report displays total sales of **$2,454,256.96**, total profit of **$202,386.39**, **10,000** orders, and an overall profit margin of **8.25%**.

LIVE DASHBOARD KPI	VALUE
Total Sales	$2,454,256.96
Total Profit	$202,386.39
Total Orders	10,000
Profit Margin	8.25%

1. Project Objective

The objective is to transform retail transaction data into useful business information. The dashboard is designed to help a viewer understand revenue, profit, order volume, product-category performance, regional results, and shipping-mode contribution through a single interactive report.

2. Problem Statement

Raw transaction records can be difficult to interpret when viewed row by row. A retail business needs a consolidated view of its performance so that stronger product areas, regional differences, and possible profitability concerns can be identified. This project addresses that need through KPI cards, comparison charts, and report-level filters.

3. Tools and Technologies

- Microsoft Power BI Desktop
- Power Query for data inspection and preparation
- DAX (Data Analysis Expressions) for calculated measures
- Power BI report visuals, including cards, bar charts, a donut chart, and a hierarchical visual

4. Dataset Description

The project uses a simulated retail sales dataset, as described in the original task report. The report contains transaction attributes used to compare sales and profit by order date, category, region, shipping mode, and product sub-category. The earlier project documentation describes 10,012 raw records and 10,000 records after its cleaning workflow. The live Power BI dashboard displays 10,000 distinct orders.

5. DATA PREPARATION AND MEASURES

5.1 Data Preparation in Power Query

The raw retail sales file was imported into Power BI Desktop. Power Query was used as the preparation stage before loading the data into the report model. The preparation checklist for this workflow includes:

- Reviewing column names and confirming that fields have suitable data types.
- Checking date fields such as Order Date and Ship Date are interpreted as dates.
- Checking numeric fields such as Sales, Profit, Quantity, and Discount.
- Reviewing missing values and data-quality indicators.
- Checking for duplicate transaction records and applying an appropriate duplicate-removal rule.
- Applying the changes with Close & Apply before creating report visuals.

5.2 DAX Measures

The dashboard uses measures to calculate the key business indicators. The following examples represent the measures used for the KPI calculations. The table name should match the actual name in the Power BI model.

MEASURE	DAX EXPRESSION
Total Sales	Total Sales = SUM('retail_sales_raw'[Sales])
Total Profit	Total Profit = SUM('retail_sales_raw'[Profit])
Total Orders	Total Orders = DISTINCTCOUNT('retail_sales_raw'[Order ID])
Profit Margin	Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)

The dashboard measures were assigned to standard Card visuals for the KPI summary. A standard Card is appropriate for a single summary value; the KPI visual is designed for an indicator with a trend axis.

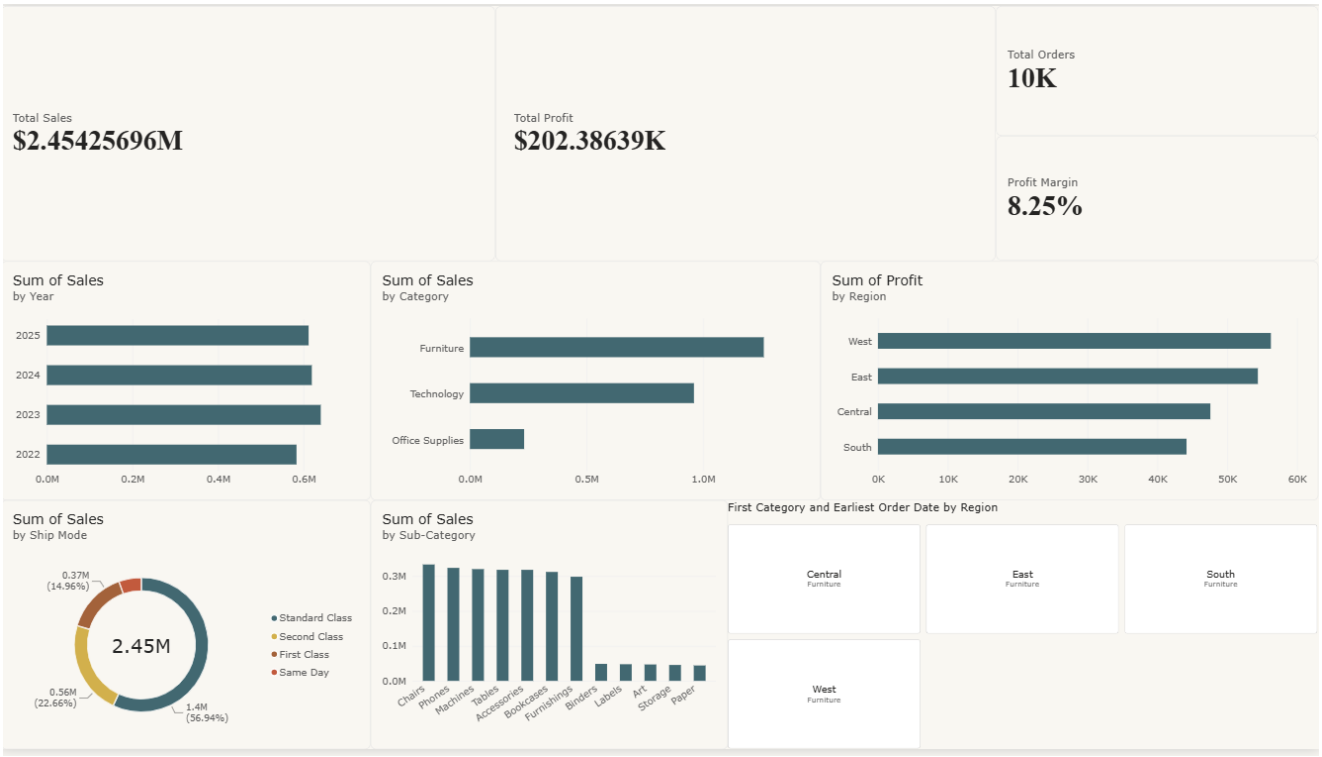
6. Dashboard Design and Components

The report page combines a KPI section with supporting charts. Its purpose is to provide a quick overview first, followed by breakdowns that help explain the headline results.

- KPI cards: Total Sales, Total Profit, Total Orders, and Profit Margin.
- Sales by Year: compares annual sales across 2022–2025.
- Sales by Category: compares Furniture, Technology, and Office Supplies.
- Profit by Region: compares West, East, Central, and South.
- Sales by Ship Mode: shows the contribution of Standard Class, Second Class, First Class, and Same Day.
- Sales by Sub-Category: compares sales across individual product sub-categories.
- Region/category hierarchical visual: presents a category-related breakdown for each region.

7. LIVE POWER BI DASHBOARD

The following image is the live Power BI Desktop dashboard screenshot supplied for this report. It is included as evidence of the current report page and its displayed values.



Dashboard observations

• Furniture is the largest sales category in the displayed category comparison, followed by Technology and Office Supplies.
• West has the highest regional profit in the displayed chart, while South has the lowest.
• Standard Class accounts for the largest share of sales by shipping mode (approximately 56.94% as labelled in the chart).
• Second Class contributes approximately 22.66%, First Class approximately 14.96%, and Same Day approximately 5.62%.
• Annual sales are shown for 2022, 2023, 2024, and 2025. The chart allows the viewer to compare yearly sales totals.

8. BUSINESS INSIGHTS

Based on the KPI cards and comparisons visible in the live dashboard, the following observations can be made:

1. The dashboard reports total sales of approximately $2.45 million.
2. Total profit is approximately $202.4 thousand.
3. The overall profit margin displayed is 8.25%.
4. The dashboard reports 10,000 orders.
5. Furniture leads the three displayed categories in sales.
6. Technology is the second-largest category by sales in the displayed chart.
7. Office Supplies has the smallest sales contribution among the three displayed categories.
8. West is the highest-profit region in the regional comparison.
9. East is the second-highest-profit region in the displayed chart.
10. South is the lowest-profit region in the regional comparison.
11. Standard Class is the dominant shipping mode by sales share.
12. Second Class is the second-largest shipping mode by sales share.
13. First Class and Same Day represent smaller shares of sales than the other two shipping modes.
14. The sub-category chart shows a noticeable gap between the leading group of sub-categories and the smaller bars at the lower end.
15. The dashboard brings sales and profitability views together, reducing the need to inspect individual transaction rows.

9. BUSINESS RECOMMENDATIONS

1. Review Furniture pricing, costs, and discounting because it has the largest sales contribution; high sales alone do not guarantee high margins.
2. Track profit alongside sales when comparing categories and regions.
3. Investigate the practices contributing to West's stronger profit result and assess whether relevant practices can be applied elsewhere.
4. Review South's regional performance to identify possible causes of its lower profit result.
5. Monitor the sales and profitability of each product sub-category to identify weaker areas.
6. Evaluate shipping-mode costs and customer requirements before changing the shipping mix.
7. Use date filters and yearly comparisons to monitor performance over time.
8. Review data quality regularly before refreshing or sharing the dashboard.
9. Keep KPI definitions consistent so sales, profit, orders, and margin are interpreted correctly.
10. Extend the report with a monthly sales trend and additional customer-level analysis when the required fields and model relationships are confirmed.

10. CHALLENGES AND LIMITATIONS

The project required the raw transaction data to be prepared before it could be used reliably in visuals. Another practical challenge was selecting the correct Power BI visual: standard Card visuals display single KPI values, whereas KPI visuals also require a trend axis. The displayed dashboard is based on simulated data, so it cannot establish the actual performance of a real retailer. The screenshot is a point-in-time view and does not demonstrate every possible filter interaction.

11. CONCLUSION

This project demonstrates a Power BI workflow for retail business intelligence: importing transaction data, preparing it for analysis, defining DAX measures, and presenting key metrics through an interactive dashboard. The completed report page provides an overview of sales, profit, order volume, margin, category performance, regional profit, shipping-mode contribution, and sub-category sales. The dashboard can be further developed with additional time-based analysis, carefully designed slicers, and validated refresh procedures.

Submission note: Save the editable Power BI report as a **.pbix** file and submit it alongside this PDF report. Because the source dataset is simulated, identify it as simulated in any presentation or written submission.